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$^{40}Ar/^{39}Ar$ incremental-release ages of hornblende and biotite from Grenville basement rocks within the Indian Head Range complex, southwest Newfoundland: their bearing on Late Proterozoic – Early Paleozoic thermal history

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Hornblende and biotite from autochthonous basement rocks within the Indian Head Range complex of southwest, insular Newfoundland record undisturbed $^{40}Ar/^{39}Ar$ release spectra with average total-gas ages of 880 Ma (hornblende) and 825 Ma (biotite). These gas-retention ages date the times when this segment of the western Appalachian basement terrane cooled below hornblende and biotite argon retention temperatures (~500°C and ~300°C respectively) following culmination of the ~1150–1100 Ma Grenville metamorphism. Although these results indicate that elevated temperatures were maintained for a prolonged period following the Grenville thermal peak, once initiated, cooling must have been relatively rapid because hornblende and biotite record generally similar total-gas dates.

The undisturbed release spectra of minerals within the Indian Head Range complex indicate that this segment of the western Appalachian basement terrane was not affected by Paleozoic metamorphism. This is consistent with recent tectonic models that indicate that the overlying Humber Arm allochthon was emplaced into its present position as a cold, already assembled structural unit. Lack of Paleozoic metamorphism within the Indian Head basement rocks is also compatible with suggestions that the obduction site of the Bay of Islands ophiolite lay considerably east of the Early Paleozoic continental margin of North America.

La hornblende et la biotite des roches autochtones du socle dans le complexe d'Indian Head Range au sud-ouest de Terre-Neuve enregistrent les spectres non influencés de libération de $^{40}Ar/^{39}Ar$ avec des âges de gaz total moyens de 880 Ma (hornblende) et de 825 Ma (biotite). Ces âges de rétention des gaz datent de l'époque où ce segment de socle à l'ouest des Appalaches s'est refroidi en dessous des températures de rétention de l'argon dans la hornblende et la biotite (~500°C et ~300°C respectivement) à la suite de la culmination, il y a ~1150–1100 Ma, du métamorphisme du Grenville. Bien que ces résultats indiquent que les températures élevées se sont maintenues pour une période de temps prolongée à la suite du maximum thermique du Grenville, une fois amorcé, le refroidissement doit avoir été relativement rapide parce que la hornblende et la biotite enregistrent généralement des dates semblables pour le gaz total.

Le spectre de libération non influencé des minéraux dans le complexe d'Indian Head Range indique que ce segment du socle des Appalaches occidentales n'a pas été affecté par le métamorphisme du Paléozoïque. Ceci est compatible avec les modèles tectoniques récents qui indiquent que l'allochtone susjacent de Humber Arm a été mis en place dans sa position actuelle comme une unité structurale froide déjà assemblée. L'absence de métamorphisme datant du Paléozoïque dans les roches du socle d'Indian Head est aussi compatible avec les suggestions que le site d'obduction de l'ophiolite de Bay of Islands se trouvait beaucoup plus à l'est de la bordure continentale au début du Paléozoïque en Amérique du Nord.

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Introduction

A terrane of predominantly high-grade gneisses and associated catazonal plutonic rocks unconformably underlies Upper Proterozoic – Lower Paleozoic, platformal sedimentary rocks along the western margin of the Appalachian orogen. Inliers of this basement terrane intermittently occur from Newfoundland to Georgia (Fig. 1). Westernmost portions of the basement inliers were generally unaffected by Paleozoic metamorphism and record high-grade, granulite-facies assemblages. Eastern segments of the Appalachian inliers were severely retrograded by Paleozoic thermal events and are characterized by metamorphic assemblages similar in grade to those developed in cover rocks.

Rb–Sr whole-rock and U–Pb zircon dates suggest that most of the basement terrane along the western margin of the Appalachian orogen represents an extension of rocks exposed within the Grenville Province and contiguous Adirondacks (see, for example, Tilton *et al.* 1960; Davis *et al.* 1962; Pringle *et al.* 1971; Fullagar and Odom 1973). Available data indicate that localized isotopic homogenization occurred within the basement rocks concomitant with a granulite-facies metamorphism that culminated at ~1150–1100 Ma. In retrograded segments of the Appalachian basement inliers, K–Ar and Rb–Sr mineral ages range from ~450 to ~350 Ma and reflect the Paleozoic thermal disturbance of mineral isotopic systems (see, for example, Dallmeyer 1975*a*,*b*: Dallmeyer and Sutter 1976). In non-retrograded portions of the western Appalachian basement terrane, mineral ages are also markedly younger than Rb–Sr whole-rock and U–Pb zircon dates, ranging from ~1000 to ~800 Ma (see, for example, Dallmeyer 1975*a*,*b*; Dallmeyer *et al.* 1975; Dallmeyer and Sutter 1976). This discordancy was originally attributed to disturbing thermal effects of a Late Proterozoic (~800 Ma) metamorphic event (Long and Kulp 1962). However, recent petrologic and $^{40}Ar/^{39}Ar$ studies suggest, instead, that younger mineral ages are recorded in non-retrograded portions of the western Appalachian basement terrane because of slow cooling following culmination of the Grenville thermal event (Dallmeyer 1974, 1975*a*,*b*; Dallmeyer *et al.* 1975; Dallmeyer and Sutter 1976).

Previous $^{40}Ar/^{39}Ar$ studies within the western Appalachian basement terrane have been limited to portions of the Reading Prong and the southern Blue Ridge (Fig. 1). This report presents $^{40}Ar/^{39}Ar$ incremental-release ages of minerals from autochthonous basement rocks within the Indian Head Range complex in southwestern Newfoundland (Figs. 1, 2). When compared with data from central and southern Appalachian basement inliers, the new results allow for a more complete evaluation of the Late Proterozoic – Early Paleozoic thermal history of the entire western Appalachian basement terrane.

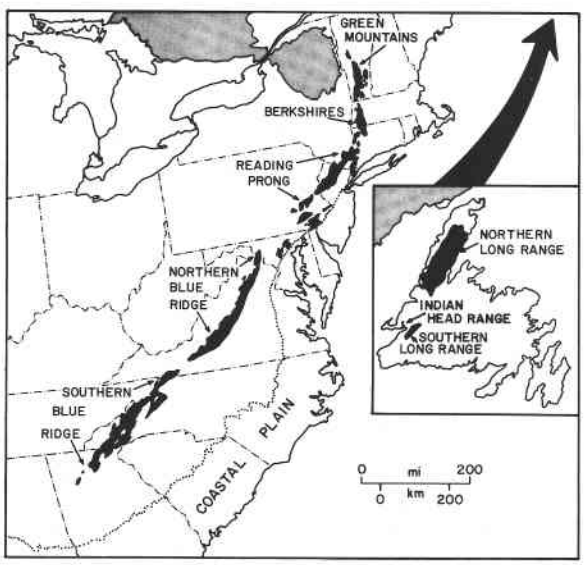


FIG. 1. Generalized map showing distribution of the western Appalachian basement terrane (black). Grenville terrane of the Canadian Shield and the Adirondacks indicated with gray shading.

Geology and Tectonic Setting of the Indian Head Range Complex

Reports of field studies in the vicinity of the Indian Head Range complex have been summarized by Riley (1962) and Williams (1975). The following description is compiled from these sources.

Basement rocks within the Indian Head Range complex display wide lithologic variation. Abundant rock types include anorthosite, granitic and tonalitic gneisses, amphibolite, hornblende–quartz–two feldspar gneisses, and various types of generally granitic pegmatites. This suite is identical to that which characterizes most exposures of the western basement terrane throughout the Appalachians. The Indian Head Range complex is exposed in two autochthonous outcrop areas at the northeastern end of St. George's Bay (Fig. 2). Each area is locally in fault or unconformable contact with platformal, Lower Cambrian – Middle Ordovician sedimentary rocks. Together, this composite autochthonous section is overlain by transported, Upper Proterozoic – Middle Ordovician sedimentary rocks within the lowermost structural unit of the Humber Arm allochthon. Transported

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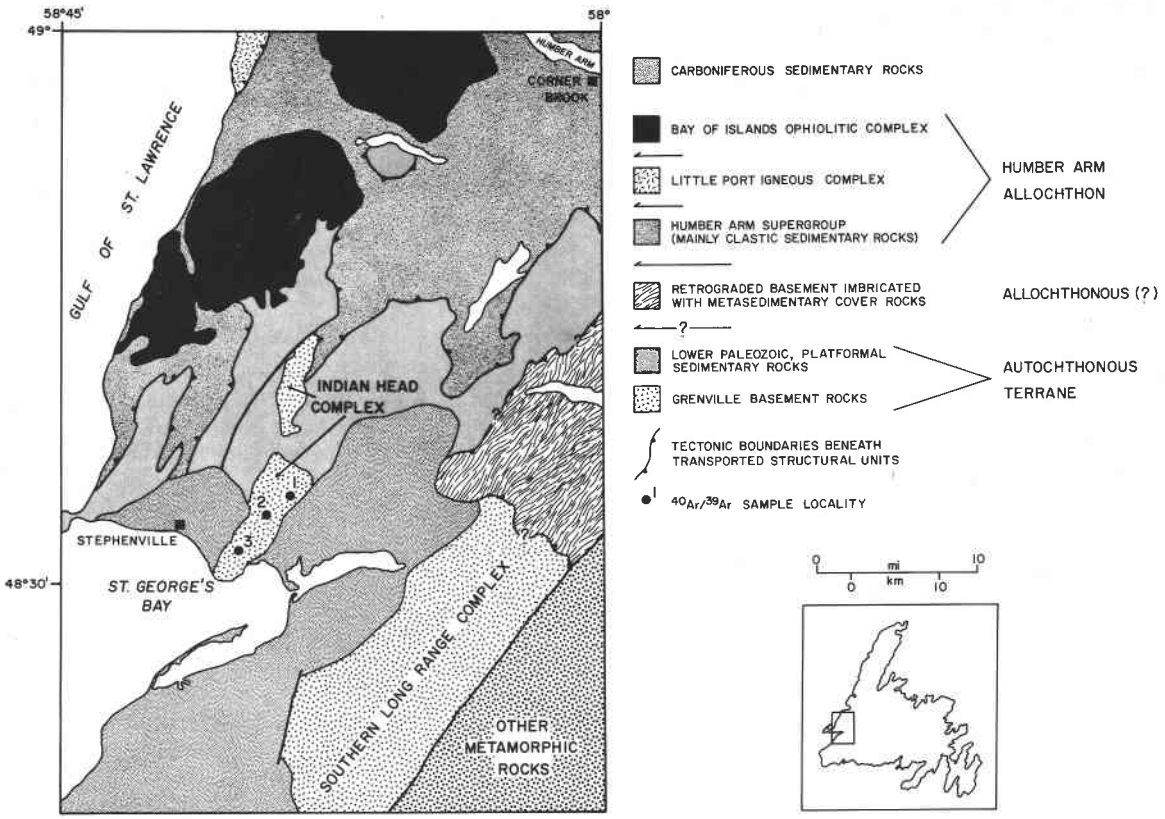


FIG. 2. Generalized geologic map of the St. George's Bay – Humber Arm region. $^{40}Ar/^{39}Ar$ sample localities within the Indian Head Range complex are indicated.

opholitic rocks within higher structural units of the allochthon are exposed in the northwestern portion of the area.

The southern outcrop area of the Indian Head Range complex is locally in fault or unconformable contact with Carboniferous sedimentary rocks. To the east, Carboniferous rocks unconformably overlie severly retrograded basement rocks which are tectonically imbricated with high-grade, Late Proterozoic – Early Paleozoic metasedimentary rocks which may be equivalent to the Fleur de Lys terrane of the White Bay – Burlington area (Harold Williams, personal communication, 1977).

Sample Location and General Petrography

Samples were collected from three localities within the southern outcrop area of the Indian Head Range complex (Fig. 2). All samples lack petrographic evidence of retrograde alteration and pure mineral concentrates were obtained.

$^{40}Ar/^{39}Ar$ Method and Analytical Techniques

The $^{40}Ar/^{39}Ar$ incremental-release dating technique has been described in detail by others (see, for example, Dallmeyer 1975*a*,*b*) and will not be reviewed. The specific types of analytical techniques employed in this study have been described elsewhere (Dallmeyer 1975*a*,*b*; Dallmeyer *et al.* 1975) and only a brief outline is presented here.

Samples of mineral concentrates were encapsulated in quartz vials and irradiated for 30 h in the United States Geological Survey TRIGA reactor (Denver). Following irradiation, gases were extracted from each sample by radiofrequency, induction heating in a series of successively higher temperature increments. Samples were maintained at each temperature increment for 1 h. The gases obtained at each temperature increment were purified and argon isotopic ratios measured with a mass spectrometer under static operating conditions. Between 8 and 12 sets of argon isotopic ratios were obtained for each temperature increment and linear zero-time extrapolations were made. Nonatmospheric ^{36}Ar and non-K-derived ^{39}Ar were corrected by monitoring ^{37}Ar and using the correction factors presented by Dalrymple and Lanphere (1971) for the reactor used in this study. Uncertainties in the calculated dates were estimated from

TABLE 1. $^{40}Ar/^{39}Ar$ analytical data for incremental heating experiments on biotite from the Indian Head Range complex, southwest Newfoundland

Release temp. (°C)	$(^{40}Ar/^{39}Ar)$*	$(^{36}Ar/^{39}Ar)$*	^{39}Ar (% of tot.)	Radiogenic (%)	Age (Ma)†
		Sample 1A, $J = 0.003203$			
525	254.30	0.48516	1.19	43.68	574±40
650	187.42	0.05578	9.50	91.21	823±20
800	179.89	0.02028	16.55	96.67	834±15
950	176.57	0.01178	11.61	98.03	831±15
1025	180.26	0.02599	6.28	95.74	830±15
1075	175.49	0.01455	11.22	98.17	825±15
1100	172.52	0.00781	17.46	98.66	820±15
Fusion	176.63	0.00814	26.19	98.69	836±15
Total	178.49	0.02205	100.00	96.35	827±15
		Sample 1B, $J = 0.003559$			
600	128.28	0.05961	2.84	86.28	626±30
800	161.05	0.01103	25.91	97.98	840±15
950	164.81	0.02477	13.33	95.56	838±15
1025	160.38	0.01043	7.77	98.09	837±15
1100	159.92	0.00570	14.20	98.95	842±15
Fusion	158.28	0.00123	35.95	99.77	841±15
Total	159.42	0.00992	100.00	98.16	835±15
		Sample 2, $J = 0.003178$			
600	225.79	0.19534	12.57	74.46	807±20
800	180.38	0.03218	36.96	94.73	818±15
950	179.95	0.03772	12.03	93.81	810±15
1025	173.87	0.02262	13.41	96.16	803±15
1100	174.96	0.01886	19.20	96.82	812±15
Fusion	184.60	0.04434	5.83	92.91	820±20
Total	184.37	0.05022	100.00	91.96	812±15

*Measured.
†Calculated using correction factors of Dalrymple and Lanphere (1971); $^{37}Ar/^{39}Ar$ ratio < 0.20 in all analyses; 2σ error estimates (rounded to 5 Ma); $\lambda = 5.303 \times 10^{-10}$ year^{-1}.

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standard deviations in zero-time extrapolations, uncertainties in the monitor age, and variations in the neutron flux (2σ, rounded to 5 Ma).

Results

$^{40}Ar/^{39}Ar$ incremental-release ages have been determined for three biotite and two hornblende concentrates. Coexisting biotite and hornblende were analyzed from locality no. 2 and hornblende from locality no. 3 (Fig. 2). Biotite concentrates from two different specimens from locality no. 1 were also analyzed. The analytical data are presented in Tables 1 and 2 and are plotted as incremental-release age spectra in Figs. 3 and 4.

The three biotite concentrates display undisturbed (concordant) release spectra (Fig. 3). They have total-gas ages of 812, 827, and 835 Ma (all ±15 Ma). These total-gas ages are mutually identical within analytical uncertainty and their average (825 Ma) is similar to a 830 Ma K–Ar age published by Lowdon (1961) for a biotite from the Indian Head Range complex.

The two hornblende concentrates analyzed in this study have undisturbed release spectra (Fig. 4). They have mutually similar total-gas ages of 884 and 878 Ma (both ±15 Ma) that are slightly older than the Indian Head biotite dates.

Interpretation

The undisturbed nature of hornblende and biotite release spectra indicate that basement rocks within the Indian Head Range complex were not affected by any Paleozoic thermal event. Instead, the data suggest that the $^{40}Ar/^{39}Ar$ total-gas ages represent gas-retention dates that record the times at which this segment of the western Appalachian basement terrane cooled below temperatures required for argon retention within hornblende and biotite (~500°C and ~300°C respectively) following culmination of the ~1150–1100 Ma Grenville metamorphism. The results indicate that elevated temperatures (>500°C) were maintained for a prolonged period (~350–300 Ma) following the Grenville thermal peak. However, once initiated, post-metamorphic cooling must have been relatively rapid because hornblende and biotite (with differ-

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TABLE 2. $^{40}Ar/^{39}Ar$ analytical data for incremental heating experiments on hornblende from the Indian Head Range complex, southwest Newfoundland

Release temp. (°C)	$(^{40}Ar/^{39}Ar)$*	$(^{36}Ar/^{39}Ar)$*	$(^{37}Ar/^{39}Ar)$†	^{39}Ar (% of tot.)	Radiogenic (%)	$^{36}Ar_{Ca}$ (%)	Age (Ma)‡
			Sample 2, $J = 0.006133$				
800	146.22	0.27202	1.976	2.52	45.19	0.21	642±25
950	106.60	0.03227	2.860	10.66	91.28	2.41	883±20
1025	101.59	0.01642	2.911	43.76	95.46	4.88	881±15
1050	98.05	0.00495	2.877	41.78	98.75	15.81	880±15
Fusion	158.44	0.20077	2.882	1.28	62.73	0.31	898±30
Total	101.82	0.01825	2.862	100.00	94.93	4.26	879±15
			Sample 3, $J = 0.006227$				
850	346.83	0.97213	2.360	4.54	17.31	0.07	600±40
1000	272.00	0.60176	3.732	1.59	34.80	0.17	875±35
1025	120.02	0.07892	5.147	10.10	80.93	1.77	894±20
1050	99.42	0.00791	4.958	62.08	98.05	17.04	896±15
1075	100.65	0.01675	4.828	15.33	95.47	7.84	887±15
Fusion	106.21	0.03240	4.932	6.36	91.37	4.14	893±15
Total	116.70	0.07123	4.818	100.00	82.22	1.84	884±15

*Measured.
†Corrected for ^{37}Ar decay.
‡Calculated using correction factors of Dalrymple and Lanphere (1971); 2σ error estimates (rounded to 5 Ma); $\lambda = 5.305 \times 10^{-10}$ year^{-1}.

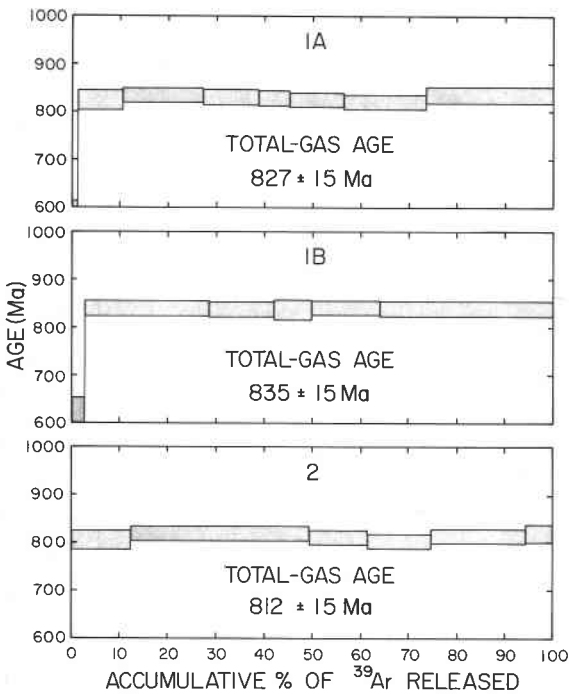


FIG. 3. $^{40}Ar/^{39}Ar$ age spectra of biotite from the Indian Head Range complex. Data plotted as age vs. accumulative percentage of ^{39}Ar released. Increasing experimental release temperatures from left to right. Estimated uncertainties (2σ, rounded to 5 Ma) indicated by width of bar. Total-gas ages listed on each spectrum.

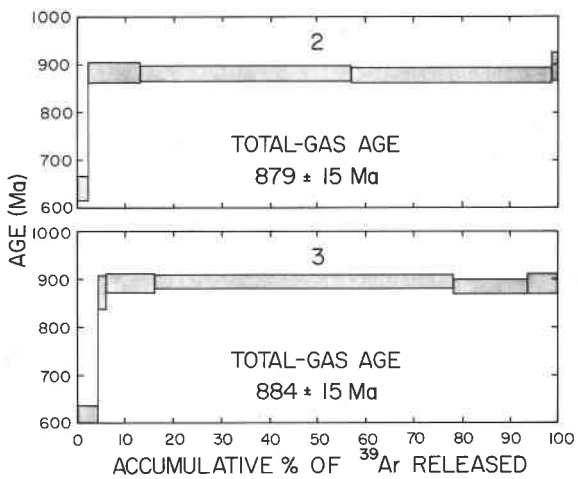


FIG. 4. $^{40}Ar/^{39}Ar$ age spectra of hornblende from the Indian Head Range complex. Data plotted as in Fig. 3.

ent temperatures of argon retention) record similar total-gas dates within the Indian Head Range complex.

The Late Proterozoic thermal history inferred for the Indian Head Range complex is different from that outlined for non-retrograded portions of Grenville basement inliers within the central and southern Appalachians (see, for example, Dallmeyer 1975*a*,*b*; Dallmeyer *et al.* 1975). In these areas, hornblende and biotite record undisturbed $^{40}Ar/^{39}Ar$ release spectra with average total-gas ages of ~975–925 Ma and ~825–775 Ma, respectively. The data appear to suggest that these segments of the western Appalachian basement terrain cooled below hornblende argon retention temperatures prior to the Indian Head Range complex rocks and that a slower rate of cooling was maintained until biotite argon retention temperatures were reached.

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Discussion

The fact that basement rocks within the Indian Head Range complex were not affected by Paleozoic thermal events is remarkable in view of the tectonic complexity of the St. George's Bay – Humber Arm area. It supports the suggestion of Williams (1975) that the thermally active obduction site for the Bay of Islands ophiolite must have been far to the east of the Lower Paleozoic continental margin of North America. It also is consistent with his proposal that the entire Humber Arm allochthon was cold and already assembled before it was emplaced into its present position by gravity sliding.

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Determination of stratigraphic thickness of strained units

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Given the state of homogeneous total strain or average total strain of a deformed unit, its stratigraphic thickness can be readily determined. The necessary theoretical development is presented herein, together with a general discussion of the necessary equations. It is shown that the practice of destraining lines of true thickness of deformed strata is theoretically sound only if the bedding surface is a principal plane of deformation.

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